

SQL Skills Diagnostic Assessment

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Pre-Study Self-Evaluation for The Science of Learning SQL

Purpose: Complete this diagnostic before beginning the 12-week learning schedule from Chapter 1. Your results will reveal exactly which foundational areas need deeper focus and which chapters to prioritize. Answer each question independently — resist the urge to look up syntax first.

How to Use This Assessment

1. Attempt every question without external help.
 2. Record your confidence level (Low / Medium / High) alongside each answer.
 3. Check your responses against the Expected Outputs section below.
 4. Use the Score Interpretation Guide to map your gaps to specific chapters.
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Section 1: Core SELECT Queries (Chapter 3)

Q1. Write a query that retrieves only the product_name and price columns from a table called products.

Q2. You notice your results contain repeated city names. Write a query that returns a unique list of cities from a customers table, with the column labelled unique_city.

Q3. What does the following query return, and what problem does it solve?

```
SELECT first_name AS "First Name", last_name AS "Last Name"  
FROM employees;
```

Section 2: Filtering With WHERE (Chapter 4)

Q4. Write a query to retrieve all orders from the orders table where total_amount is greater than 500.

Q5. Retrieve all customers whose country is 'Canada' AND whose signup_year is 2022 or later.

Q6. Write a query to find all products where the description column contains no value at all (the field is empty/absent).

Q7. Using LIKE, write a query that returns all employees from the employees table whose last_name begins with the letter 'S'.

Section 3: Sorting and Limiting (Chapter 5)

Q8. Write a query that returns the 5 most expensive products from the products table, sorted highest price first.

Q9. You want to sort sales records by region alphabetically, then by sale_amount from highest to lowest within each region. Write that query.

Q10. Place the following SQL clauses in their correct logical execution order (the order the database processes them, not the written order):

LIMIT — ORDER BY — WHERE — SELECT — FROM

Section 4: Aggregation and GROUP BY (Chapter 6)

Q11. Write a query that counts the total number of rows in the orders table.

Q12. Find the average salary for each department in an employees table. Label the result avg_salary.

Q13. Modify your Q12 query to show only departments where the average salary exceeds 60,000.

Q14. Explain in one sentence why COUNT(column_name) and COUNT(*) can return different numbers.

Section 5: Joins (Chapter 7)

Q15. Write an INNER JOIN query that combines orders and customers on customer_id, returning customer_name and order_total.

Q16. You want every customer to appear in results, even those with no orders. Which join type do you use, and why?

Q17. What happens if you forget the ON condition in a JOIN statement?

Expected Outputs & Model Answers

Q#	Key Concept Tested	Model Answer / Expected Output
Q1	Column selection	SELECT product_name, price FROM products;
Q2	DISTINCT + alias	SELECT DISTINCT city AS unique_city FROM customers;
Q3	Column aliasing	Returns employee names with readable headers; solves unclear column labels
Q4	Comparison operator	WHERE total_amount > 500
Q5	AND + date filter	WHERE country = 'Canada' AND signup_year >= 2022
Q6	NULL handling	WHERE description IS NULL
Q7	LIKE wildcard	WHERE last_name LIKE 'S%'
Q8	ORDER BY + LIMIT	ORDER BY price DESC LIMIT 5
Q9	Multi-column sort	ORDER BY region ASC, sale_amount DESC
Q10	Execution order	FROM → WHERE → SELECT → ORDER BY → LIMIT
Q11	COUNT aggregate	SELECT COUNT(*) FROM orders;
Q12	GROUP BY + AVG	SELECT department, AVG(salary) AS avg_salary FROM employees GROUP BY department;
Q13	HAVING clause	Add HAVING AVG(salary) > 60000
Q14	NULL counting trap	COUNT(column_name) skips NULL values; COUNT(*) counts every row
Q15	INNER JOIN	JOIN customers ON orders.customer_id = customers.customer_id
Q16	LEFT JOIN	LEFT JOIN — preserves all left-table rows regardless of matches
Q17	Cartesian product	Rows multiply uncontrollably, silently corrupting results

Score Interpretation Guide

- 14–17 correct: Strong foundation — use the 12-week schedule as designed.
- 10–13 correct: Targeted gaps — revisit flagged chapters before advancing.
- Below 10: Begin at Chapter 3 and apply the spaced repetition system from Chapter 1 systematically.

Remember: A low score here is diagnostic data, not failure. The cognitive science in Chapter 1 shows that identifying gaps before studying is precisely how active recall accelerates durable learning.